Daisyworld with an Ocean: a guide to exploring the app

EC2113 Intro to Climate · Ocean module · app: joymonteiro.github.io/intro_climate_science/tutorials/daisyworld-ocean

This handout goes with the app "Daisyworld with an Ocean". Open the app next to it. The app runs in any browser, including on a phone, and needs no installation.

- **Part 1** explains what the model is and how to use the app.
- **Part 2** walks through six runs. For each one it says what to set, asks you to predict what will happen, and then explains what you see. Try to answer each "Before you look" question before you read on.
- **Part 3** sums up what to take away, and says what the model leaves out.
- **An extension** at the end adds daisies and grazers, if you have time.

You need no maths beyond the lecture's slab equation.

> **The point in two sentences.** Weather pushes on every surface of a planet, and the surface responds according to how much heat it can lean on. A deep, stirred ocean takes out the fast swings and turns weather into slow wandering, but it doesn't change the size of the slowest swings: **depth sets the timescale, not the size of the slow swings.**

Part 1. The model and the app

1.1 What is in the model, in words

The first three presets are a bare planet: no life, just physics.

- **Land.** One temperature for all the land. It stores little heat, so it responds within weeks.
- **A mixed layer.** One temperature for the stirred top of the ocean, 75 m deep, covering 70% of the planet. It has about 30 times the land's heat capacity.
- **A deep ocean.** One temperature for all the water below. It holds 13 times as much heat as the mixed layer and exchanges heat with it slowly.
- **The sun.** Sunlight rises and sets every day, and through the year swings 30% above and below its average, highest on 21 June (the **Day and seasons** box, ticked in presets 1 to 3).
- **The weather.** A random push of energy, in or out, that lasts a few days, like a weather system passing. A day's push has a spread of 30 W/m². Nothing in the weather swings slowly. The **same push** falls on land and ocean.
- **Heat loss to space.** A warmer surface radiates more. Every part of the model that touches the air feels this pull back toward balance.

There are no places, no map and no currents. **No water moves sideways.** All the ocean does here is store heat, and mixing up and down decides how much of it the surface can use.

1.2 The lecture's equation, and the app's numbers

Each part of the planet obeys the slab equation from the lecture:

$C\,dT'/dt = F'(t) - \lambda T'$

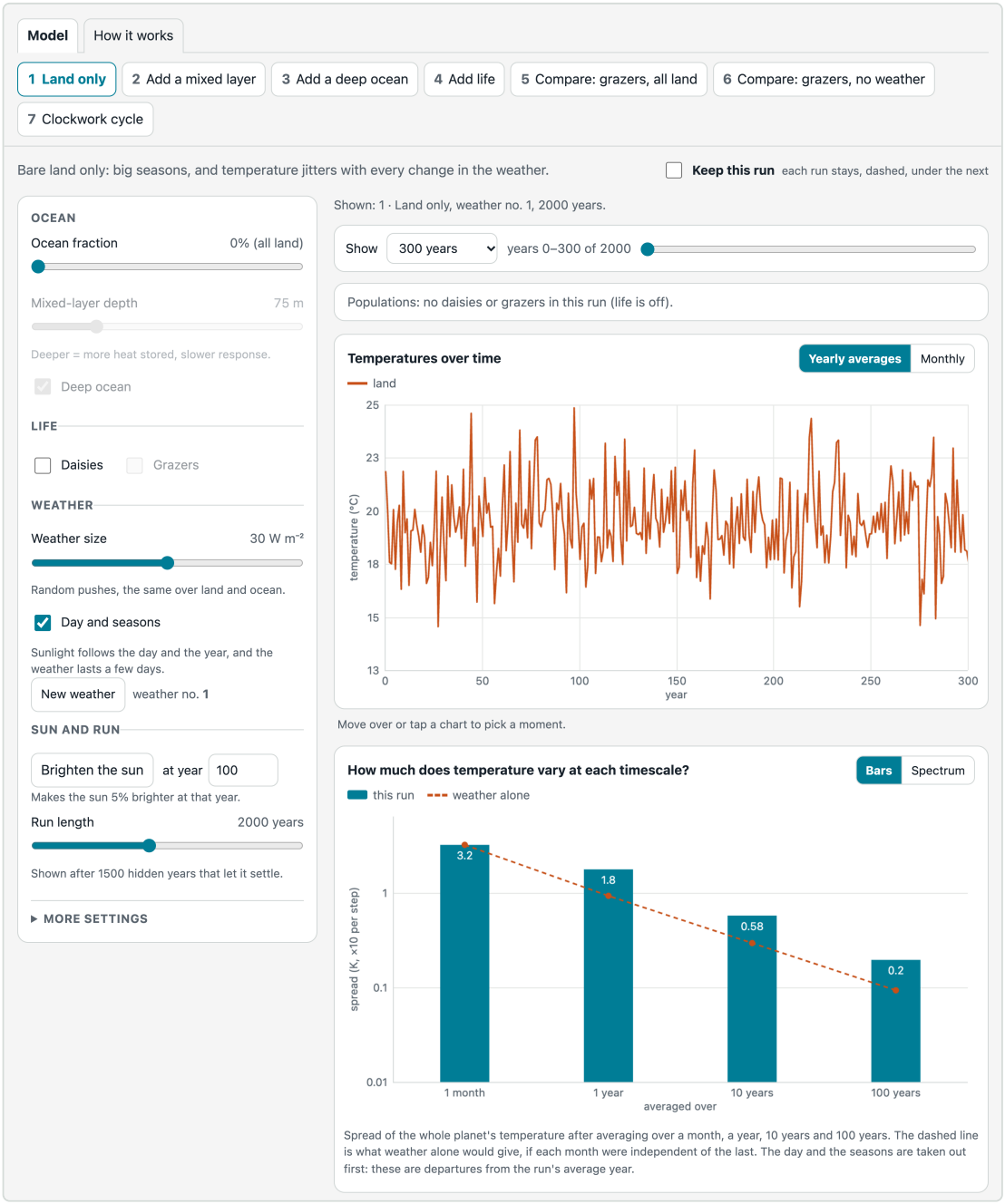
Here C is the heat capacity, F′ is the weather and λ says how fast a warmer surface loses heat. The response time is $\tau = C/\lambda$.

The app's numbers differ a little from the lecture's, for good reasons:

	Lecture	App	Why they differ
Mixed-layer depth	50 m	75 m	Both are typical. You can change it with a slider.
λ	3.8 W m^{-2} K^{-1}	2.09 W m^{-2} K^{-1}	The lecture's value is the Planck response alone. The app's includes other feedbacks, such as water vapour, so a warm surface loses heat more slowly.
Land	about 1 m of water	2.4 m of water	The app's land box carries the air above it. The lecture's estimate is the soil alone.
Mixed layer τ	~1.7 years	~5 years	Both of the differences above.
Land τ	~12 days	~2 months	The same.

The physics is the same in both. Don't worry about the exact numbers; watch the patterns.

1.3 A tour of the screen



- **Presets** run along the top. Use 1 to 3 for this handout: **Land only**, **Add a mixed layer**, **Add a deep ocean**. A preset runs by itself as soon as you press it.
- **Keep this run** keeps the current run on the charts, dashed, so you can compare the next run with it. Tick it before you load the next preset.
- **Temperatures over time** shows each part of the planet. The switch gives yearly averages or monthly values. To see the seasons, press **Monthly** and show 10 years.
- **How much does temperature vary at each timescale?** has two views. **Bars** shows the spread of the planet's temperature after averaging over a month, a year, 10 years and 100 years. **Spectrum** shows the same thing in finer detail. On both, a dashed line shows what weather alone would give, if each month were independent of the last. Both take the seasons out first: they use each month's departure from the run's average month of its kind (its average January, and so on), because the seasons repeat every year and would swamp everything else.
- **New weather** runs again with a different random sequence. Use it to check that something isn't luck.

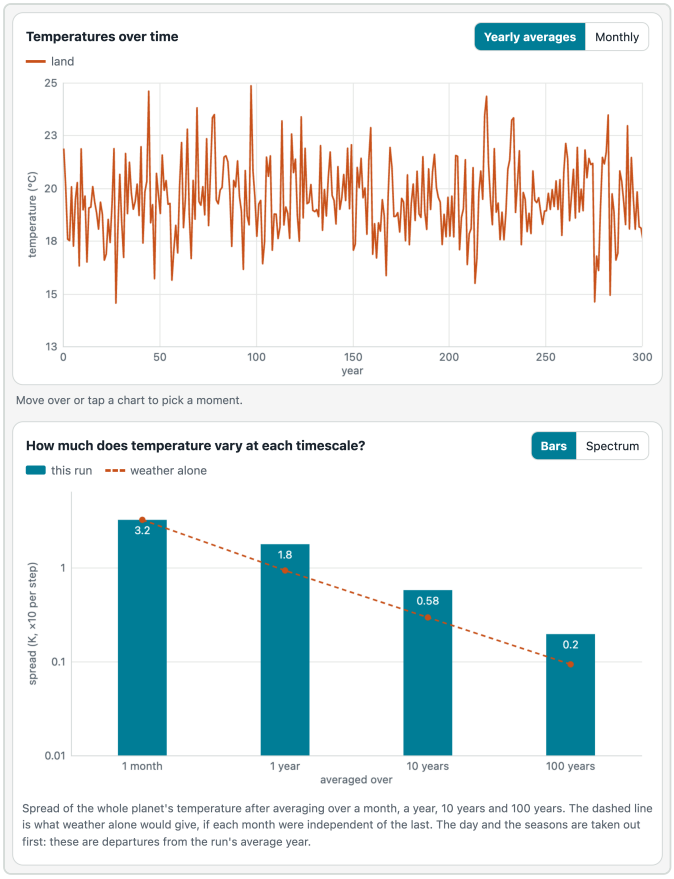
The "How it works" tab explains the model in plain language.

Part 2. Six runs to explore

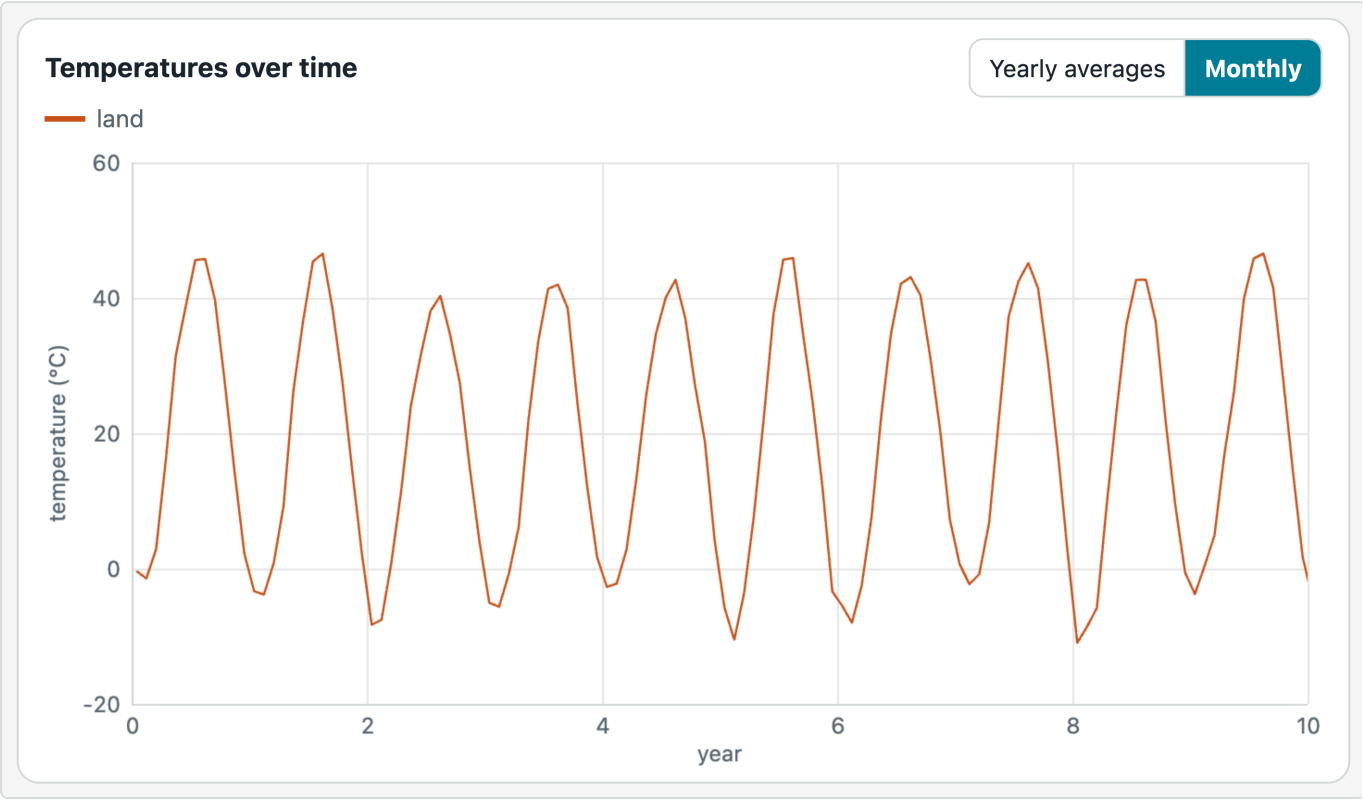
Run 1. Land only

Set: preset 1, **Land only**.

Before you look: the weather is random, and forgets within a few days. What should a planet of bare land do? Sketch the temperature you expect over 300 years. If you average over a whole year, then 10 years, how fast should the wiggles shrink?



What to look for. First press **Monthly** and show 10 years: the seasons are huge, from about −5 °C in February to 44 °C in August.



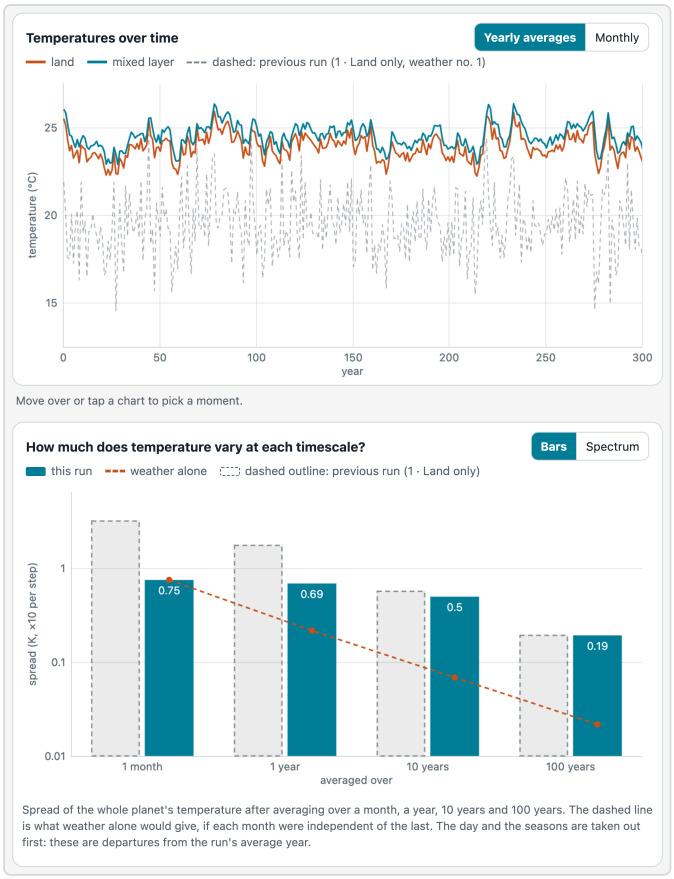
Then go back to **Yearly averages**: the temperature jitters with no pattern you can see. Each bar is smaller than the last: about 3.2 K for monthly values, 1.8 K for yearly averages, 0.6 K over 10 years and about 0.2 K over 100 years.

Why. Land forgets within about two months. After that, each push has faded, so averaging over longer periods mostly cancels pushes against each other. The bars fall close to the dashed line. The seasons are big for the same reason: land holds so little heat that it nearly catches up with the sun each season.

Run 2. Add a mixed layer

Set: tick **Keep this run**, then preset 2, **Add a mixed layer**. Same weather.

Before you look: you are about to give 70% of the planet 30 times the heat capacity. What happens to the seasons? Will the planet vary more, less or the same from **month to month**? From **century to century**? Write down your answers.

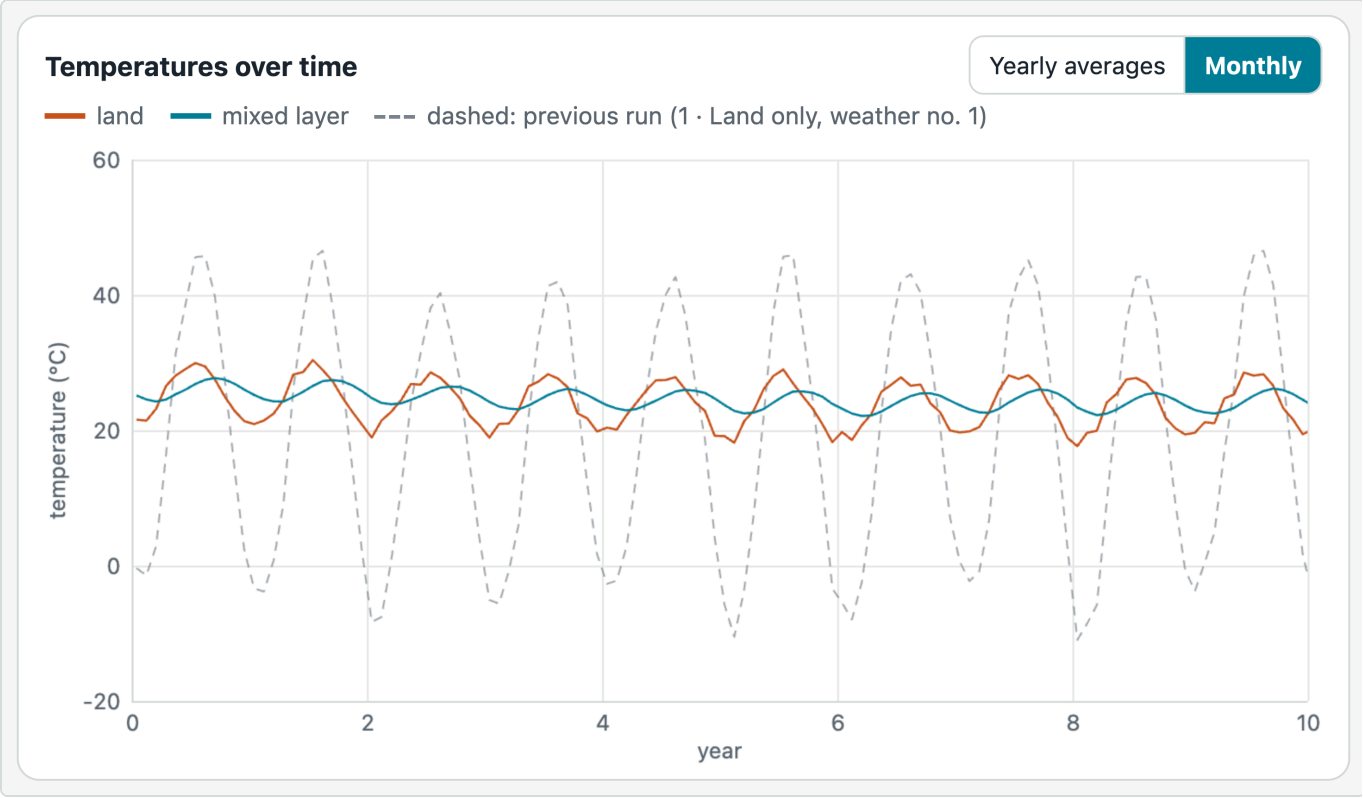


What to look for.

Spread of the planet's temperature	Land only	With a mixed layer
over 1 month	3.20–3.26 K	0.74–0.80 K
over 1 year	1.75–1.83 K	0.67–0.74 K
over 10 years	0.55–0.67 K	0.45–0.54 K
over 100 years	0.14–0.28 K	0.14–0.26 K

(Ranges cover weather numbers 1 to 20. Your numbers should fall in or near them.)

The fast bars shrink a lot and the century bar doesn't change. Now look at the yearly bar next to the monthly one: with the mixed layer, it hardly drops. On **Monthly**, the seasons shrink too: the mixed layer swings only about 3 K through the year, warmest in September, nearly three months after the sun, and the land next to it about 9 K.



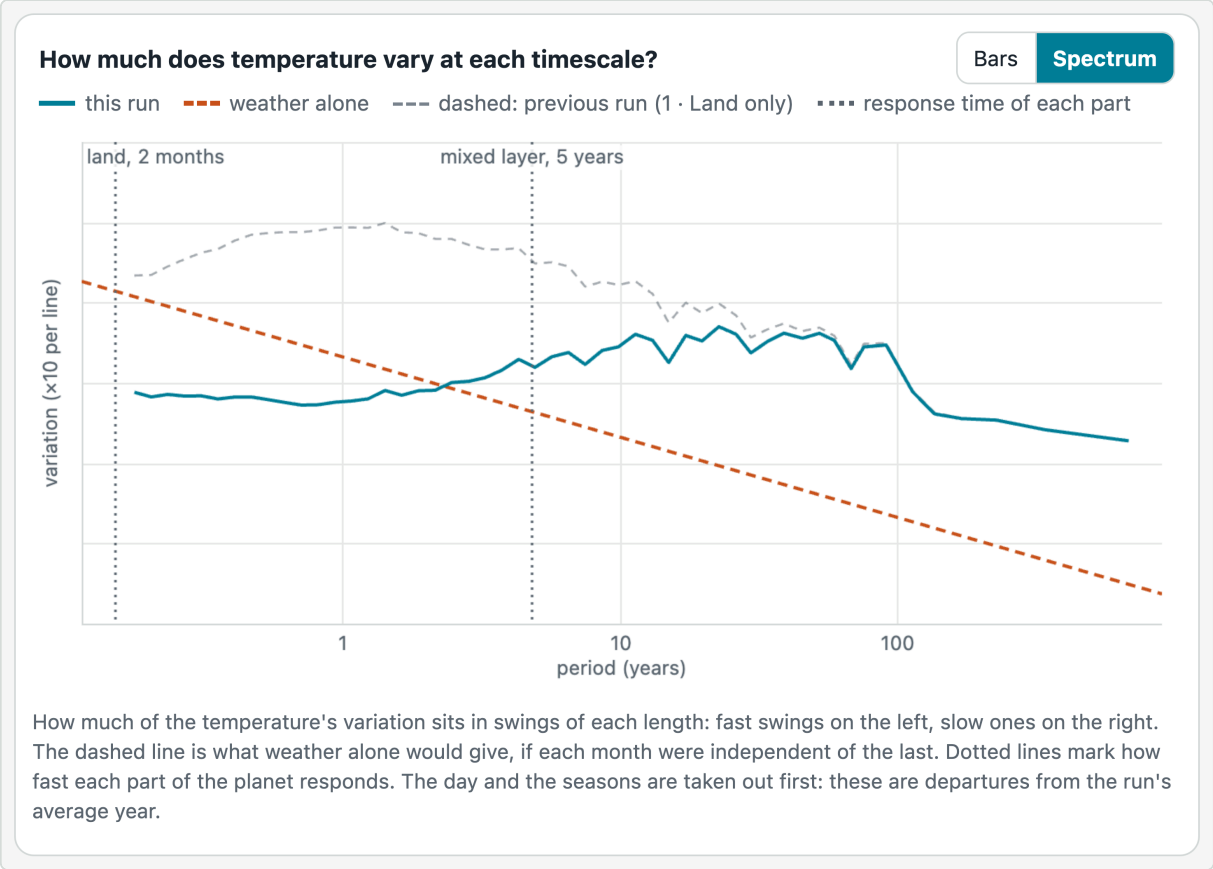
Why. A month of warm weather warms the mixed layer only a little, but it hasn't forgotten that warmth when the next month comes. It adds up the kicks. So its temperature barely registers one month, then wanders over years. Over a century, what matters is only how strongly heat is lost to space, and that didn't change. A season is a slow push too, but not slow enough for a layer that takes about 5 years to respond: it never catches up, so its seasons are small and late.

The ocean doesn't calm the planet evenly. It **moves** variation from the fast end to the slow end.

Run 3. Sort it by timescale

Set: stay on the same two runs and press **Spectrum**.

Before you look: the curve for the land-only run and the curve for the mixed layer should be far apart somewhere and together somewhere else. On which side of the chart will each happen?



How much of the temperature's variation sits in swings of each length: fast swings on the left, slow ones on the right. The dashed line is what weather alone would give, if each month were independent of the last. Dotted lines mark how fast each part of the planet responds. The day and the seasons are taken out first: these are departures from the run's average year.

What to look for. On the left, at periods of months to a couple of years, the mixed-layer curve sits about **100 times** below land's (each gridline is a factor of 10). On the right, past about 30 to 50 years, the two curves lie on top of each other.

Why. For fast swings, the push doesn't have time to be balanced by heat loss, so C alone sets the response. For slow swings, the surface has time to reach balance, $F' \approx \lambda T'$, and C drops out of the equation.

Two details. The dotted lines mark each part's response time τ. A curve bends at a period of about $2\pi \times \tau$, because a swing of period P varies at a rate $\omega = 2\pi/P$, so the bends sit somewhat to the right of the dotted lines. And this chart shows the temperature of the whole planet, which is 30% land. The mixed layer on its own would be several hundred times below land.

Run 4. Change the depth

Set: preset 2, then move **Mixed-layer depth** to 25 m. Tick **Keep this run**, then move it to 150 m.

Before you look: a mixed layer six times deeper. What happens to the month bar? What happens to the century bar?

What to look for.

Spread of the planet's temperature	25 m	75 m	150 m
over 1 year	1.08–1.14 K	0.67–0.74 K	0.48–0.54 K
over 10 years	0.51–0.62 K	0.45–0.54 K	0.37–0.45 K
over 100 years	0.14–0.28 K	0.14–0.26 K	0.13–0.24 K

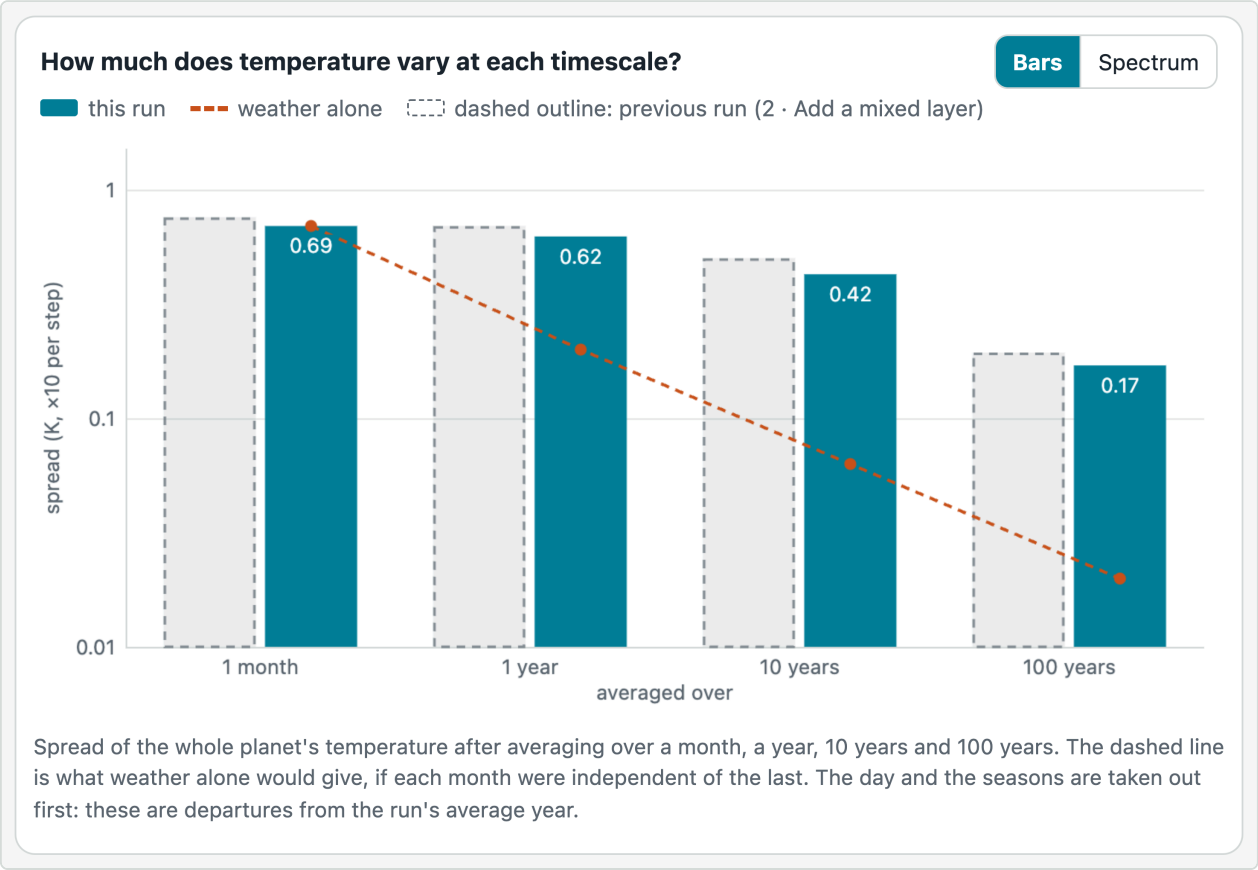
Why. This is the main result. **Depth sets the timescale, not the size of the slow swings.** A deeper layer holds on for longer, so the fast swings shrink and the wander gets slower. The size of the slowest swings stays put, because λ sets it, and λ doesn't care how deep the water is.

Think about it: depth here means how far the stirring reaches. The monsoon stirs the Arabian Sea from about 20 m in April to about 60 m in July. What does that do to the sea's memory during the year?

Run 5. Add a deep ocean

Set: preset 2, tick **Keep this run**, then preset 3, **Add a deep ocean**.

Before you look: a deep ocean holds 13 times more heat than the mixed layer. Do you expect more slow variation or less?



Spread of the whole planet's temperature after averaging over a month, a year, 10 years and 100 years. The dashed line is what weather alone would give, if each month were independent of the last. The day and the seasons are taken out first: these are departures from the run's average year.

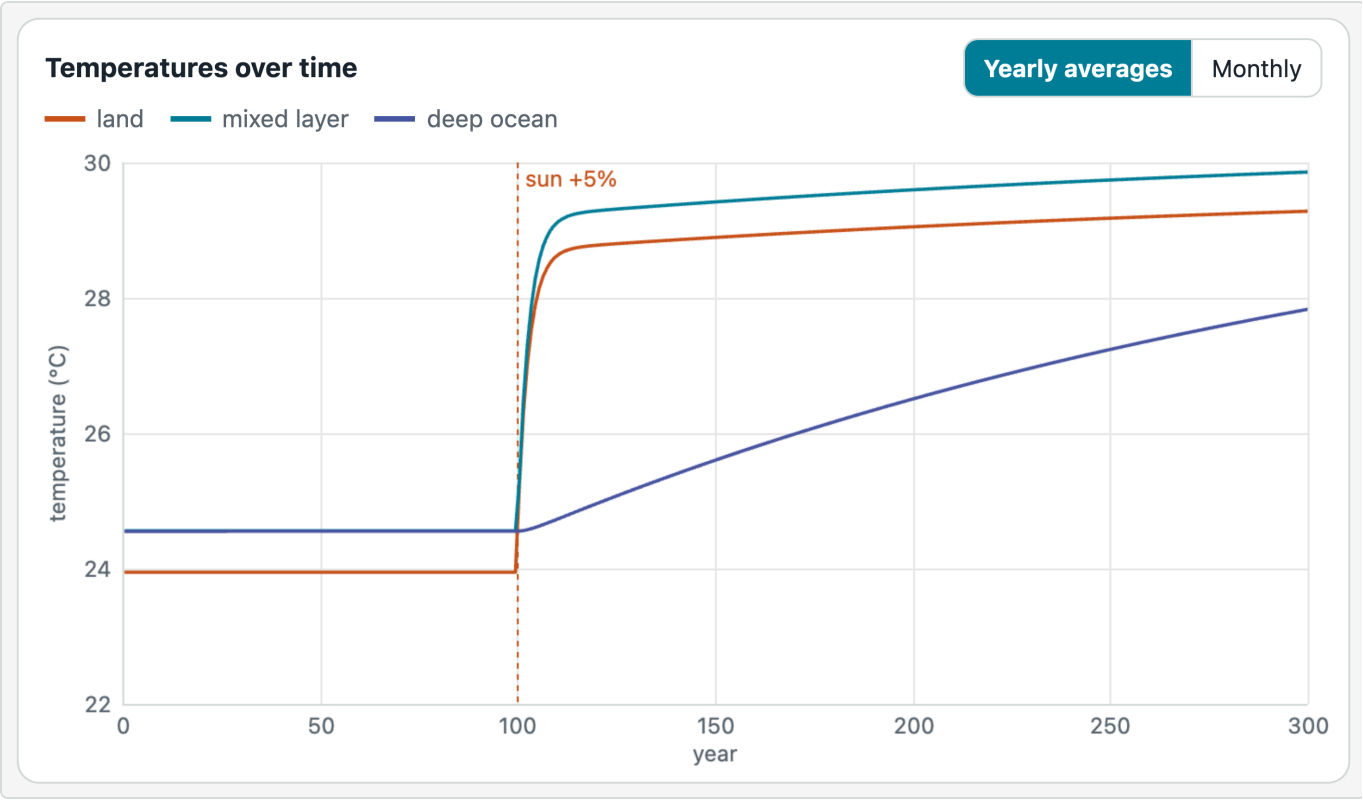
What to look for. The century bar **falls**, from 0.14–0.26 K to 0.12–0.22 K. On the spectrum, power at 100 to 1,000 years falls by about a third.

Why. Many people expect a slower part to add slow variation. Here it soaks up heat instead: when the mixed layer wanders warm, some of that warmth leaks down, and the leak acts like extra heat loss on long timescales. The deep ocean calms the slowest swings a little.

Run 6. Brighten the sun

Set: preset 3. Move **Weather size** to 0, then press **Brighten the sun**. Note what happens on **Yearly averages**, which hide the seasons. Then untick **Deep ocean** and do it again.

Before you look: the sun brightens by 5% and stays there. How long until the planet stops warming? A year? Ten? Does the deep ocean change your answer?



What to look for.

Share of the final warming (5.8 K) reached after	1 year	10 years	100 years	500 years
With a deep ocean	35%	80%	87%	98%
Without a deep ocean	37%	95%	100%	100%

The planet warms in **two stages**: most of it within 10 years, and the last fifth over centuries.

Why. The mixed layer catches up within a few of its response times. But it keeps leaking heat into the deep ocean, which takes centuries to fill. **This is the shape of the warming we are living through**: more than 91% of Earth's extra heat is going into the ocean, and the surface hasn't finished responding to today's imbalance.

Part 3. What to take away

1. **How much heat a surface can lean on depends on how heat gets below it.** Land has only conduction, so its slab is always thin. The ocean is stirred, so its slab is tens to hundreds of metres.
2. **The ocean adds up weather.** Fast, random pushes become slow wandering, with nothing slow driving it.
3. **Depth sets the timescale, not the size of the slow swings.**
4. **A deep ocean adds a slower stage.** It calms the slowest swings a little, and it makes warming arrive in two stages.
5. **All of this happens with no water moving sideways.** Mixing up and down is enough.

What the model leaves out. Currents, so heat can't be carried anywhere. Geography: its day and seasons are one place's, every day lasts 12 hours, and its land is one block sized for the year's swing, so it swings only about 1 K between day and night (real bare ground, warmed only in its top 10 cm or so by a day, swings far more). Any change of mixed-layer depth through the year. Feedbacks that change with the ocean's temperature, such as those behind El Niño, which is why real ocean swings can be stronger than this model's.

Questions to take to the discussion.

- Whose temperature next month would you rather forecast, Delhi's or the Arabian Sea's? Why, and what would make you less sure?
- The Arabian Sea cools from about 30 °C in May to about 27 °C in August, while the sun is still high. What in the lectures could explain that, and what in this model couldn't?

Extension. Add life

Set: preset 4, **Add life**, then **Compare: grazers, no weather**.

Daisies now set how much sunlight the land reflects, and grazers eat the daisies. The life presets run without day and seasons, with weather that forgets from one quarter-day to the next, as they were designed. For a fair comparison, load preset 3 and untick **Day and seasons**: its century bar is 0.05–0.08 K. Now load preset 4 and look at the century bar: it is two or three times as tall (0.11–0.28 K). With no weather at all, the same planet cycles every 8 years. Nothing outside is cycling: the ocean's delay makes the daisies and grazers overshoot, and the weather then decides how deep each bust goes. Find three busts of different sizes in the grazer chart and read the ocean's temperature just before each. What do you notice?